

Element A: Presentation and Justification of the Problem.

Power cables often come unplugged accidentally for many reasons, like getting kicked or pulled from their socket unintentionally. This affects the general public. The goal of our project is to prevent cables from becoming unplugged by accidentally getting kicked or pulled on. This will help people prevent loss of computer data, loss of aborted creations when 3D printing, or loss of electricity to refrigerators and freezers.

Our primary stakeholders are people with a lot of electronics, electricians, Ms. Z, who is a science teacher at our school, student Z, office workers, and gamers. They are stakeholders because they may have many electronics that could get unplugged. Mr. K, who is an engineering teacher at our school, deals with cables being unplugged unintentionally when he is at his mom's house because the sockets are loose.

Cables or cords can get unplugged when they get tripped over, when moving things on your desk, or when yanking game controllers. This can happen at homes, offices, and schools, which can interrupt or delete important work. This happens more often in high-traffic areas and areas with lots of movement.

Solving this problem will be more convenient for people and prevent accidental data loss. We also believe that this will improve work efficiency since people won't have to stop working when a cable gets unplugged. Solving this problem will have a large impact because this happens so frequently.

When we did research, we found out that the prongs in the outlet that hold the electrical plug in get worn out over time. The quality of the outlet matters because a lower quality outlet becomes looser quicker (Director, 2025). As buildings get older, so does the electrical wiring in them, which also means the outlets deteriorate and get

looser.

References

Director, J. B.-M. (2025, September 8). Why are outlets loose, and how to fix them.

The Electricians.

<https://www.theelectriciansco.com/blog/why-your-outlets-are-loose/>